Nandan Kamath

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EDUCATION

University of Michigan, Ann Arbor, MI **August 2023 - May 2027**

BSE, Computer Engineering

- Relevant Coursework: Very Large Scale Integrated Design (VLSI) I, Parallel Computer Architecture, Computer Architecture, Digital Integrated Circuits, Computer Organization, Logic Design, Signals and Systems, Data Structures and Algorithms

TECHNICAL PROJECTS

R10K Style Out-of-Order Processor **August 2025 - December 2025**

- Implemented an N-way superscalar, out-of-order CPU inspired by the MIPS R10K microarchitecture in System Verilog, operating at 83.33 MHz with dynamic scheduling, physical register renaming, speculative execution, and in-order retirement
- Created a centralized Reorder Buffer, Reservation Station, Free List, map tables, and pipelined variable-latency functional units, supporting full branch misprediction recovery and control-flow optimization using a GShare branch predictor
- Optimized memory access by designing a non-blocking instruction cache with stream buffer prefetching and a 4-way set-associative, write-back data cache with MSHRs for miss handling and memory ordering through the Store Queue

VLSI Design & Optimization of a 16-Bit RISC Processor **January 2026 - April 2026**

- Engineered the physical layout of a 16-bit RISC processor in Cadence Virtuoso, designing transistor-level schematics and layouts for the ALU, Logarithmic Shifter, Register File, Decoder, and Program Counter that pass DRC/LVS requirements
- Optimized ALU critical path performance by implementing a 16-bit Kogge-Stone parallel prefix adder, leveraging $O(\log_2 N)$ carry logic to achieve a 23% reduction in delay (avg. 272 ps improvement in HSPICE) and a 32% reduction in silicon area.
- Validated system functionality using NC-Verilog, performing mixed-signal verification to ensure the processor's logic between the behavioral model and the extracted Cadence transistor netlist was equivalent across all instruction types

Hardware Acceleration for Real-time Semantic Graph SLAM **February 2026 - April 2026**

- Developed custom acceleration kernels using the AMD AIE library to map matrix multiplications to an AMD Versal chip, utilizing its AI engines to overcome programmable logic FPGA space constraints and integrating FP32 to INT8 quantization.
- Profiled and accelerated the compute-intensive convolutional layers of the RangeNet++ architecture, optimizing the underlying matrix calculations to enable real-time semantic segmentation of 2D projections of 3D LiDAR point clouds.
- Benchmarked and simulated performance, comparing the Versal architecture against a baseline AWS EC2 g5.xlarge instance (NVIDIA A10 GPU), achieving an estimated 82% reduction in execution latency and a 90% improvement in power efficiency.

CPL and CMOS Adder Design **October 2025 -November 2025**

- Created two full-custom 8-bit dual-mode ripple-carry adders in Cadence Virtuoso using complementary pass logic for high speed and static CMOS for low power, achieving a 2 GHz clock speed and 151 µW performance targets, respectively.
- Optimized device sizing, critical-path, and node capacitances across >180 µm of transistor width, integrating transmission gate registers and pass-transistor multiplexers to meet timing and functional correctness in adder and accumulation modes.
- Verified functionality and timing through Cadence transient simulations and custom-designed testbenches, validating correct circuit operation across all mode transitions and measuring power/delay tradeoffs between the two architectures.

Sequential Four-Function Calculator **December 2024 - January 2025**

- Implemented a sequential four-function calculator in Verilog on the Altera DE2-115 FPGA board, enabling addition, subtraction, multiplication, and division of 11-bit signed integers with real-time input via slider switches and push buttons
- Developed a finite state machine controller and integrated an arithmetic datapath incorporating a custom adder, subtractor, Booth's algorithm multiplier, and repeated subtraction division modules, with accurate operation and overflow detection
- Simulated and debugged calculator operations in ModelSim by verifying controller state transitions and datapath relations, and ensured stable 50MHz operation by resolving timing issues and handling edge cases like overflow and divide-by-zero

PROFESSIONAL EXPERIENCE

Walmart Inc. **Bentonville, AR**

Automation Engineering Intern **June 2025 - August 2025**

- Conducted data analysis using SQL and Google Cloud Platform (GCP) to identify throughput constraints, revealing that optimizing the tote consolidation process would add 100,000 cubic feet of storage, equivalent to a 2% gain in volume
- Built an Integer Linear Programming (ILP) optimization tool in Python to assign associates to workstations based on volume targets, availability, and productivity, minimizing headcount while consistently achieving 94–106% of outbound demand
- Oversaw installation and commissioning of next-generation robotic consolidation system across multiple sites, used data analysis to estimate merge throughput, and determined that 22 robots would be required to meet demand through FY31

SKILLS

- Computer Programming: C/C++, SystemVerilog, Python, TCL, Linux, SQL
- Engineering Tools: Synopsys (Verdi, Design Compiler), Cadence (Virtuoso, JasperGold, Innovus), ModelSim, HSPICE
- VLSI/Architecture: CPU & GPU Architecture, Processor Pipelines, Memory Hierarchy, Power and Performance, NoC, Cache Coherency, Synthesis & P&R, PPA Optimization, DRC/LVS, UVM, ARM/RISC-V ISA, ASIC/FPGA, RTL, Verification, Networking